

HOPSTEN, Germany

WMO: 103140

Lat: 52.333N

Lon: 7.533E

Elev: 40

StdP: 100.85

Time Zone: 1.00 (EUC)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -7.0	(d) -11.6	(e) 1.4	(f) -8.7	(g) -9.2	(h) 1.7	(i) -5.5	(j) 13.9	(k) 8.1	(l) 12.3	(m) 8.0	(n) 2.8	(o) 70	(p) 0.549

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	30.4	19.5	28.2	18.5	26.0	17.5	20.6	27.9	19.4	25.8	18.3	24.2	3.0	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	13.0	22.8	17.0	12.2	21.9	16.0	11.5	20.7	59.3	28.0	55.1	25.8	51.9	24.5	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 10.0	(b) 8.6	(c) 7.6	(e) -12.4	(f) 33.7	(g) 4.3	(h) 1.5	(i) -15.5	(j) 34.8	(k) -18.0	(l) 35.6	(m) -20.4	(n) 36.5	(o) -23.5	(p) 37.6		
(a) 10.0	(b) 8.6	(c) 7.6	(e) -12.5	(f) 22.4	(g) 4.2	(h) 1.1	(i) -15.6	(j) 23.2	(k) -18.0	(l) 23.8	(m) -20.4	(n) 24.4	(o) -23.4	(p) 25.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	2.7	3.6	6.1	8.9	13.4	15.5	18.1	18.0	14.3	10.4	5.7	2.6		
	DBStd	6.72	4.69	4.69	3.19	3.88	3.88	3.36	3.25	3.12	2.71	3.55	3.54	4.48		
	HDD10.0	1018	228	182	126	66	12	1	0	0	1	41	132	231		
	HDD18.3	3148	485	413	379	283	161	98	44	44	127	247	379	487		
	CDD10.0	1007	1	2	5	33	116	166	250	249	128	52	3	2		
	CDD18.3	95	0	0	0	1	6	13	36	35	4	0	0	0		
	CDH23.3	1150	0	0	0	10	98	178	419	400	44	1	0	0		
	CDH26.7	359	0	0	0	0	17	51	139	142	10	0	0	0		
(14) Wind	WSAvg	3.3	4.1	4.1	3.8	3.3	3.0	2.9	2.7	2.6	2.8	3.2	3.3	3.6		
(15) Precipitation	PrecAvg	831	75	63	61	48	58	69	82	74	77	69	74	76		
	PrecMax	1061	133	134	129	96	128	130	168	135	156	218	125	174		
	PrecMin	650	7	20	13	4	30	26	33	20	9	16	26	25		
	PrecStd	128	39	32	32	22	24	29	37	31	38	43	30	35		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.1	18.1	25.0	28.9	31.3	32.9	32.9	28.0	22.1	14.8	13.2		
		MCWB	10.3	10.2	11.2	14.5	17.6	20.6	21.0	19.8	17.8	16.4	11.6	11.2		
	2%	DB	11.2	12.7	15.1	21.0	26.0	27.9	30.2	30.3	23.8	19.1	13.0	12.0		
		MCWB	9.0	9.1	10.0	13.4	16.4	18.5	19.9	19.3	17.1	14.9	11.4	10.5		
	5%	DB	10.1	11.1	13.0	18.3	23.8	25.0	27.9	27.9	20.8	17.1	11.7	10.2		
		MCWB	8.2	8.6	9.2	11.7	15.4	17.5	18.9	18.5	15.7	14.1	9.8	8.6		
	10%	DB	8.7	9.6	11.3	15.7	21.1	22.6	25.3	25.1	19.0	15.5	10.2	8.8		
		MCWB	7.2	7.5	8.4	10.6	14.3	16.3	17.5	17.7	15.0	13.0	8.7	7.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	10.8	12.1	15.7	18.1	21.3	21.9	22.0	19.3	17.0	12.7	11.9		
		MCDB	12.5	13.6	16.1	22.4	26.6	28.6	31.3	30.2	26.5	20.3	13.8	12.9		
	2%	WB	9.5	9.8	10.8	14.2	17.1	19.5	20.7	20.4	17.5	15.5	11.4	10.4		
		MCDB	10.8	12.0	13.8	19.6	24.2	26.2	28.4	27.4	22.1	18.2	12.8	11.9		
	5%	WB	8.4	8.7	9.8	12.7	15.8	18.0	19.4	19.1	16.5	14.3	10.1	8.8		
		MCDB	9.9	10.8	12.3	16.9	22.1	23.2	25.9	25.3	20.0	16.6	11.3	10.1		
	10%	WB	6.9	7.6	8.9	11.2	14.7	16.7	18.2	18.2	15.5	13.2	8.9	7.3		
		MCDB	8.3	9.4	11.2	14.9	20.1	21.5	23.9	24.1	18.3	15.3	10.1	8.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	6.1	7.5	9.7	10.7	10.3	10.8	11.3	9.1	7.8	5.8	4.7		
		MCDBR	6.3	7.6	10.5	13.5	14.9	15.0	15.5	16.2	12.2	9.7	6.9	5.9		
	5% WB	MCWBR	5.2	5.5	6.7	7.2	7.0	7.4	6.8	7.1	6.8	6.3	5.3	5.0		
		MCWBR	5.7	6.8	8.8	11.8	13.0	13.3	13.9	13.6	10.4	9.2	6.5	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.346	0.376	0.410	0.412	0.410	0.422	0.415	0.389	0.361	0.341	0.319		
		taud	2.357	2.340	2.269	2.218	2.247	2.271	2.248	2.287	2.350	2.409	2.373	2.342		
	Ebn at Noon		663	759	806	822	840	844	825	809	786	734	636	598		
		Edh at Noon	61	82	107	126	129	127	128	117	98	76	60	53		
(44) All-Sky Solar Radiation	RadAvg		0.68	1.35	2.48	4.02	4.89	5.20	4.98	4.18	3.04	1.77	0.81	0.53		
	RadStd		0.08	0.24	0.34	0.43	0.44	0.40	0.63	0.35	0.35	0.17	0.09	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (68 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page